

Please cite: X. Mougeot et al., Metrologia 62 (2025) 029001  
<https://dx.doi.org/10.1088/1681-7575/adb275>



1 Decay Scheme

<sup>103</sup>Pd disintegrates by almost 100% electron-capture decay via the 39,753-keV metastable state (<sup>103m</sup>Rh) to the ground state of <sup>103</sup>Rh. There is no evidence of an EC transition directly to the ground state of <sup>103</sup>Rh.  
*Le <sup>103</sup>Pd se désintègre à presque 100% par capture électronique vers le niveau fondamental du <sup>103</sup>Rh via le niveau excité de 39,753 keV. Aucune transition directe vers le niveau fondamental n'a été observée.*

2 Nuclear Data

$T_{1/2}({}^{103}\text{Pd})$  : 17,00 (5) d  
 $T_{1/2}({}^{103\text{m}}\text{Rh})$  : 56,115 (16) min  
 $Q^+({}^{103}\text{Pd})$  : 574,7 (24) keV

2.1 Electron Capture Transitions

|                  | Energy<br>(keV) | Probability<br>(%) | Nature                   | lg <i>ft</i> | <i>P<sub>K</sub></i> | <i>P<sub>L</sub></i> | <i>P<sub>M</sub></i> |
|------------------|-----------------|--------------------|--------------------------|--------------|----------------------|----------------------|----------------------|
| ε <sub>0,5</sub> | 37,9 (24)       | 0,00480 (16)       | allowed                  | 7,33         | 0,55 (5)             | 0,35 (4)             | 0,084 (11)           |
| ε <sub>0,4</sub> | 217,3 (24)      | 0,0300 (9)         | 1st forbidden non-unique | 8,569        | 0,8459 (6)           | 0,1226 (4)           | 0,02589 (17)         |
| ε <sub>0,3</sub> | 279,7 (24)      | 0,00058 (7)        | 1st forbidden non-unique | 10,52        | 0,8518 (5)           | 0,11804 (26)         | 0,02479 (14)         |
| ε <sub>0,1</sub> | 534,9 (24)      | 99,9646 (9)        | allowed                  | 5,880        | 0,86086 (30)         | 0,11103 (15)         | 0,02310 (11)         |

2.2 Gamma Transitions and Internal Conversion Coefficients

|                       | Energy<br>(keV) | P <sub>γ+ce</sub><br>(%) | Multipolarity | α <sub>K</sub> | α <sub>L</sub> | α <sub>M</sub> | α <sub>T</sub> |
|-----------------------|-----------------|--------------------------|---------------|----------------|----------------|----------------|----------------|
| γ <sub>1,0</sub> (Rh) | 39,753 (6)      | 99,9695 (12)             | E3+M4         | 142 (7)        | 1051 (30)      | 214 (6)        | 1440 (40)      |
| γ <sub>2,1</sub> (Rh) | 53,283 (6)      | 0,0000190 (12)           | (M1)          | 1,81 (3)       | 0,223 (4)      | 0,0415 (6)     | 2,08 (3)       |
| γ <sub>4,3</sub> (Rh) | 62,41 (3)       | 0,00282 (10)             | M1            | 1,143 (16)     | 0,1406 (20)    | 0,0262 (4)     | 1,314 (19)     |
| γ <sub>5,3</sub> (Rh) | 241,875 (10)    | 0,00000086 (8)           | E1            | 0,01033 (15)   | 0,001202 (17)  | 0,000222 (4)   | 0,01179 (17)   |
| γ <sub>3,0</sub> (Rh) | 294,962 (15)    | 0,00340 (11)             | M1+E2         | 0,01669 (24)   | 0,00199 (3)    | 0,000370 (6)   | 0,0191 (3)     |
| γ <sub>4,1</sub> (Rh) | 317,72 (5)      | 0,0000175 (17)           | E1            | 0,00493 (7)    | 0,000571 (8)   | 0,0001055 (15) | 0,00563 (8)    |
| γ <sub>4,0</sub> (Rh) | 357,38 (8)      | 0,0271 (8)               | E2            | 0,01368 (20)   | 0,00181 (3)    | 0,000337 (5)   | 0,01588 (23)   |
| γ <sub>5,2</sub> (Rh) | 443,809 (10)    | 0,0000189 (12)           | E2            | 0,00698 (10)   | 0,000889 (13)  | 0,0001655 (24) | 0,00807 (12)   |
| γ <sub>5,1</sub> (Rh) | 497,083 (7)     | 0,00477 (16)             | M1+E2         | 0,00458 (7)    | 0,000539 (8)   | 0,0001001 (15) | 0,00524 (8)    |

3 Atomic Data

3.1 Rh

|                 |   |        |      |
|-----------------|---|--------|------|
| ω <sub>K</sub>  | : | 0,809  | (4)  |
| ω̄ <sub>L</sub> | : | 0,0494 | (12) |
| n <sub>KL</sub> | : | 0,987  | (4)  |

3.1.1 X Radiations

|                   | Energy<br>(keV) | Relative<br>probability |                    |
|-------------------|-----------------|-------------------------|--------------------|
| X <sub>Ktot</sub> | 20,073 - 23,173 | 182,8                   |                    |
| Kα <sub>2</sub>   | 20,073          | 52,75                   | } Kα               |
| Kα <sub>1</sub>   | 20,215          | 100                     |                    |
| Kβ <sub>3</sub>   | 22,699          | 8,51                    | } K'β <sub>1</sub> |
| Kβ <sub>1</sub>   | 22,725          | 16,56                   |                    |
| KN                | 23,167 - 23,173 | 4,96                    |                    |
| X <sub>Ltot</sub> | 2,378 - 3,094   | 21,8                    |                    |
| X <sub>Mtot</sub> | 0,188 - 0,553   | 0,61                    |                    |
| X <sub>Ntot</sub> | 0,029 - 0,063   | 0,0472                  |                    |

**3.1.2 Auger Electrons**

|                        | Energy<br>(keV) | Relative<br>probability |
|------------------------|-----------------|-------------------------|
| Auger K <sub>tot</sub> | 16,281 - 22,838 | 142,8                   |
| KLL                    | 16,281 - 17,083 | 100                     |
| KLX                    | 19,133 - 20,146 | 39,09                   |
| KXY                    | 21,922 - 22,838 | 3,69                    |
| Auger L <sub>tot</sub> | 0,004 - 3,260   | 1512                    |
| Auger M <sub>tot</sub> | 0,016 – 0,394   | 3985                    |
| Auger N <sub>tot</sub> | 0,001 – 0,058   | 4302                    |

**4 Electron Emissions**

|                     |      | Energy<br>(keV) | Electrons<br>(per 100 disint.) |
|---------------------|------|-----------------|--------------------------------|
| eAN <sub>tot</sub>  | (Rh) | 0,001 - 0,058   | 555 (5)                        |
| eAM <sub>tot</sub>  | (Rh) | 0,016 - 0,394   | 514 (5)                        |
| eAL <sub>tot</sub>  | (Rh) | 0,004 - 3,260   | 195,0 (20)                     |
| eAK <sub>tot</sub>  | (Rh) | 16,281 - 22,838 | 18,42 (9)                      |
|                     | KLL  | 16,281 - 17,083 | 12,90 (6)                      |
|                     | KLX  | 19,133 - 20,146 | 5,043 (24)                     |
|                     | KXY  | 21,922 - 22,838 | 0,4760 (23)                    |
| ec <sub>1,0</sub> T | (Rh) | 16,533 - 39,751 | 99,9695 (12)                   |
| ec <sub>1,0</sub> K | (Rh) | 16,533 (6)      | 9,9 (6)                        |
| ec <sub>1,0</sub> L | (Rh) | 36,341 - 36,749 | 73 (3)                         |
| ec <sub>1,0</sub> M | (Rh) | 39,126 - 39,446 | 14,9 (6)                       |
| ec <sub>1,0</sub> N | (Rh) | 39,672 - 39,751 | 2,2 (1)                        |

## 5 Photon Emissions

### 5.1 X-Ray Emissions

|                   |      | Energy<br>(keV) | Photons<br>(per 100 disint.) |   |              |
|-------------------|------|-----------------|------------------------------|---|--------------|
| XN <sub>tot</sub> | (Rh) | 0,029 - 0,063   | 0,02001 (14)                 |   |              |
| XM <sub>tot</sub> | (Rh) | 0,188 - 0,553   | 0,2595 (23)                  |   |              |
| XL <sub>tot</sub> | (Rh) | 2,378 - 3,094   | 9,24 (10)                    |   |              |
| XK <sub>tot</sub> | (Rh) | 20,073 - 23,173 | 77,5 (4)                     |   |              |
| XK $\alpha_2$     | (Rh) | 20,073          | 22,36 (11)                   | } | K $\alpha$   |
| XK $\alpha_1$     | (Rh) | 20,215          | 42,39 (20)                   |   |              |
| XK $\beta_3$      | (Rh) | 22,699          | 3,606 (17)                   | } | K' $\beta_1$ |
| XK $\beta_1$      | (Rh) | 22,725          | 7,021 (33)                   |   |              |
| XKN               | (Rh) | 23,167 - 23,173 | 2,104 (10)                   |   |              |

### 5.2 Gamma Emissions

|                           | Energy<br>(keV) | Photons<br>(per 100 disint.) |
|---------------------------|-----------------|------------------------------|
| $\gamma_{1,0}(\text{Rh})$ | 39,753 (6)      | 0,0694 (19)                  |
| $\gamma_{2,1}(\text{Rh})$ | 53,283 (6)      | 0,0000062 (4)                |
| $\gamma_{4,3}(\text{Rh})$ | 62,41 (3)       | 0,00122 (4)                  |
| $\gamma_{5,3}(\text{Rh})$ | 241,875 (10)    | 0,00000085 (8)               |
| $\gamma_{3,0}(\text{Rh})$ | 294,962 (15)    | 0,00334 (11)                 |
| $\gamma_{4,1}(\text{Rh})$ | 317,72 (5)      | 0,0000174 (17)               |
| $\gamma_{4,0}(\text{Rh})$ | 357,38 (8)      | 0,0267 (8)                   |
| $\gamma_{5,2}(\text{Rh})$ | 443,809 (10)    | 0,0000187 (12)               |
| $\gamma_{5,1}(\text{Rh})$ | 497,083 (7)     | 0,00475 (16)                 |

## 6 Main Production Modes

<sup>102</sup>Pd(n,γ)<sup>103</sup>Pd  
 natural-Pd(n,xn)<sup>103</sup>Pd  
<sup>103</sup>Rh(p,n)<sup>103</sup>Pd  
<sup>107</sup>Ag(p,αn)<sup>103</sup>Pd  
<sup>102</sup>Pd(d,p)<sup>103</sup>Pd  
<sup>103</sup>Rh(d,2n)<sup>103</sup>Pd  
<sup>107</sup>Ag(d,α2n)<sup>103</sup>Pd  
<sup>103</sup>Rh(α,p3n)<sup>103</sup>Pd  
 natural-Ag(p,2pxn)<sup>103</sup>Pd  
 natural-In(p,4pxn)<sup>103</sup>Pd

## 7 References

- A. FLAMMERSFELD. *Naturwissenschaften* 32 (1944) 36-37  
(Rh-103m half-life)
- M.L. WIEDENBECK. *Phys. Rev.* 67 (1945) 267-268  
(Rh-103m half-life)
- M.L. WIEDENBECK. *Phys. Rev.* 68 (1945) 237-239  
(Rh-103m half-life)
- A. FLAMMERSFELD, O. BRUNA. *Z. Naturforsch.* 2a (1947) 241-244  
(Rh-103m half-life)
- D.E. MATTHEWS, M.L. POOL. *Phys. Rev.* 72 (1947) 163-164  
(Pd-103 half-life)
- M. LINDNER, I. PERLMAN. *Phys. Rev.* 73 (1948) 1202-1203  
(Pd-103 half-life)
- E. KONDAIAH. *Phys. Rev.* 79 (1950) 891-892  
(40-keV K/(L+M) ratio)
- J.Y. MEI, C.M. HUDDLESTON, A.C.G. MITCHELL. *Phys. Rev.* 79 (1950) 429-432  
(Rh-103m half-life)
- J.M. CORK, J.M. LEBLANC, F.B. STUMPF, W.H. NESTOR. *Phys. Rev.* 86 (1952) 575-576  
(K/L ratios)
- W.W. MEINKE. *Phys. Rev.* 90 (1953) 410-412  
(Pd-103 half-life)
- L.H.TH. RIETJENS, H.J. VAN DEN BOLD, P.M. ENDT. *Physica* 20 (1954) 107-114  
(Pd-103 half-life, gamma-ray energies and emission probabilities)
- P. AVIGNON, A. MICHALOWICZ, R. BOUCHEZ. *J. Phys. Radium* 16 (1955) 404-410  
(Gamma-ray energies and emission probabilities, ICC(K), ICC(L), K/L ratio)
- G.M. DRABKIN, V.I. ORLOV, L.I. RUSINOV. *Izv. Akad. Nauk SSSR, Ser. Fiz., Columbia Tech. Transl.* 19 (1955) 294-304  
(ICC(K), K/L ratio, L/M ratio)
- F.K. MCGOWAN, P.H. STELSON. *Phys. Rev.* 99 (1955) 127-134  
(Mixing ratio (295 keV))
- B. SARAF. *Phys. Rev.* 97 (1955) 715-720  
(Gamma-ray energies and emission probabilities)
- G.A. JONES, W.R. PHILLIPS. *Proc. Royal Soc. (London)* 239A (1957) 487-493  
(Rh-103m half-life)
- F.K. MCGOWAN, P.H. STELSON. *Phys. Rev.* 109 (1958) 901-916  
(Mixing ratio (295 keV))
- A.M. BRESESTI, M. BRESESTI, H. NEUMANN. *J. Inorg. Nucl. Chem.* 29 (1967) 15-20  
(Ru-103 beta decay: K X-ray emission probabilities, K/(L+M+N) ratio (39.75 keV))

- A. VUORINEN. Proc. Symp. Standardization of Radionuclides, IAEA, Vienna, Austria (1967)  
(Rh-103m half-life, K/(K+L+M) ratio, K X-ray emission probabilities (39.75 keV))
- R.S. HAGER, E.C. SELTZER. Nucl. Data A4 (1968) 1-235  
(Theoretical ICC)
- J.C. MANTHURUTHIL, H.J. HENNECKE, C.R. COTHERN. Phys. Rev. 165 (1968) 1363-1370  
(ICC, multipolarities, mixing ratios)
- J.A. PANONTIN, N.T. PORILE, A.A. CARETTO JR. Phys. Rev. 165 (1968) 1273-1281  
(Pd-103 half-life)
- Y. GRUNDITZ, S. ANTMAN, H. PETTERSSON, M. SARACENO. Nucl. Phys. A133 (1969) 369-384  
(Pd-103 half-life, gamma-ray energies and emission probabilities, ICC(K), subshell ratios, K X-ray emission probabilities, (P(357 keV)/P(K X-ray)))
- Y. KOBAYASHI. JAERI report JAERI-1178 (1969) 21-22  
(Rh-103m half-life)
- M.A. LEPRI, W.S. LYON. Int. J. Appl. Radiat. Isot. 20 (1969) 297-298  
(K(alpha)/K(beta), 39.75-keV ICC(K))
- D.E. RAESIDE, J.J. REIDY, M.L. WIEDENBECK. Nucl. Phys. A134 (1969) 347-352  
(Gamma-ray energies, ICC(K))
- W.H. ZOLLER, E.S. MACIAS, M.B. PERKAL, W.B. WALTERS. Nucl. Phys. A130 (1969) 293-304  
(Gamma-ray energies and emission probabilities, 497-keV ICC(K), gamma-gamma coincidence)
- E.B. NIESCHMIDT, D.A. PEARSON. IN-1317 (1970) 122-126  
(Gamma-ray energies and emission probabilities, ICC(K), L subshell ratios)
- H. PETTERSSON, S. ANTMAN, Y. GRUNDITZ. Z. Phys. 233 (1970) 260-274  
(Gamma-ray energies, L1/L2, L1/L3, L2/L3 ratios, ICC(K))
- H. PETTERSSON, S. ANTMAN, Y. GRUNDITZ. Erratum, Z. Phys. 235 (1970) 485  
(ICC(K))
- W.M. RONEY, H.W. KUGEL, G.M. HEESTAND, R.R. BORCHERS, R. KALISH. Proc. Int. Conf. Nuclear Reactions Induced by Heavy Ions, 15-18 July 1969, Heidelberg, Germany (1970) 419-423  
(Mixing ratio (295 keV))
- F.T. AVIGNONE III, G.D. FREY. Phys. Rev. C 4 (1971) 912-918  
(ICC(K), K/(L+M+ ...) ratio, multipolarity)
- W. BAMBYNEK, B. CRASEMANN, R.W. FINK, H.-U. FREUND, H. MARK, C.D. SWIFT, R.E. PRICE, P. VENUGOPALA RAO. Rev. Mod. Phys. 44 (1972) 716-813  
(K-shell fluorescence yield)
- D.S. BRENNER, M.L. PERLMAN. Nucl. Phys. A181 (1972) 207-216  
(39.75-keV L1/L3, L2/L3 ratios)
- H.C. JAIN, S.K. BHATTACHERJEE, C.V.K. BABA. Nucl. Phys. A178 (1972) 437-448  
(Half-life of 93-keV nuclear level of Rh-103)
- A. PAZSIT, J. CSIKAI. Sov. J. Nucl. Phys. 35 (1972) 232-233  
(Rh-103m half-life, 39.75-keV ICC(K))
- R.O. SAYER, J.K. TEMPERLEY, D. ECCLESHELL. Nucl. Phys. A179 (1972) 122-140  
(Mixing ratio (295 keV))
- CHR. BARGHOLTZ, J. BECKER, L. ERIKSSON, L. GIDEFELDT, L. HOLMBERG, V. STEFANSSON. Physica Scripta 8 (1973) 90-94  
(Half-life of 93-keV nuclear level of Rh-103)
- E. GÜNTHER, K. KNAUF, K.F. WALZ. Int. J. Appl. Radiat. Isot. 24 (1973) 87-89  
(Rh-103m half-life)
- H. ING, W.G. CROSS. Int. J. Appl. Radiat. Isot. 24 (1973) 437-450  
(Ru-103 beta decay: K X-ray emission probabilities)
- W. BAMBYNEK, B. CRASEMANN, R.W. FINK, H.-U. FREUND, H. MARK, C.D. SWIFT, R.E. PRICE, P. VENUGOPALA RAO. Erratum, Rev. Mod. Phys. 46 (1974) 853  
(K-shell fluorescence yield)
- K.H. CZOCK, N. HASELBERGER, F. REICHEL. IAEA report IAEA/RL/26 (1974)  
(ICC(total), ICC(K), K/(L+M+ N) ratio)
- D.C. SANTRY, J.P. BUTLER. Can. J. Phys. 52 (1974) 1421-1428  
(Rh-103m half-life, K X-ray emission probabilities)
- K.H. CZOCK, N. HASELBERGER, F. REICHEL. Int. J. Appl. Radiat. Isot. 26 (1975) 417-421  
(ICC(total), ICC(K), K X-ray emission probabilities)
- K.H. CZOCK, N. HASELBERGER, F. REICHEL, S. POPA. Int. J. Appl. Radiat. Isot. 26 (1975) 782-784  
(Pd-103 half-life)

- B. MARTIN, D. MERKERT, J.L. CAMPBELL. Z. Phys. A274 (1975) 15-21  
(39.75-keV L1/L2, L1/L3, L2/L3, M, N, O ratios)
- E.S. MACIAS, M.E. PHELPS, D.G. SARANTITES, R.A. MEYER. Phys. Rev. C 14 (1976) 639-644  
(Gamma-ray energies and emission probabilities, ICC(K), ICC(L), K X-ray emission probabilities, (P(357 keV)/P(K X-ray)) ratio)
- K.S. KRANE. At. Data Nucl. Data Tables 19 (1977) 363-416  
(Mixing ratios)
- F.P. LARKINS. At. Data Nucl. Data Tables 20 (1977) 311-387  
(Auger-electron energies)
- F. LAGOUTINE, J. LEGRAND, C. BAC. Int. J. Appl. Radiat. Isot. 29 (1978) 269-272  
(Rh-103m half-life)
- D. NEWTON, A.E.R. TOUREAU, A.L. ANDERSON, R.A. MEYER. Int. J. Appl. Radiat. Isot. 29 (1978) 188-189  
(Gamma-ray energies and emission probabilities, (P(375 keV)/P(K X-ray)) ratio)
- R. VANINBROUKX, G. GROSSE, W. ZEHNER. NEA report NEANDC(E)-202U Vol. III (1979) 28-30  
(Rh-103m gamma-ray emission probability, K X-ray emission probabilities, ICC(total), ICC(K))
- E. HAGN, J. WESE, G. ESKA. Z. Phys. - Atoms and Nuclei A299 (1981) 353-358  
(Mixing ratio (497.1 keV))
- D.W. MURRAY, A.L. ALLSOP, N.J. STONE. Hyperfine Interactions 11 (1981) 127-139  
(Mixing ratio (497.2 keV))
- R. VANINBROUKX, G. GROSSE, W. ZEHNER. Int. J. Appl. Radiat. Isot. 32 (1981) 589-591  
(Pd-103 and Rh-103m half-lives)
- R. VANINBROUKX, W. ZEHNER. Int. J. Appl. Radiat. Isot. 32 (1981) 850  
(Rh-103m K X-ray emission probabilities)
- K.S. KRANE. Phys. Rev. C 27 (1983) 411-412  
(Mixing ratio (497.1keV))
- S.T. PERKINS, D.E. CULLEN, M.H. CHEN, J.H. HUBBELL, J. RATHKOPF, J. SCOFIELD. Lawrence Livermore National Laboratory report UCRL-50400 30 (1991)  
(Atomic database)
- D.E. CULLEN, S.T. PERKINS, S.M. SELTZER. Appl. Radiat. Isot. 44 (1993) 1343-1347  
(Atomic database)
- U. SCHÖTZIG. Appl. Radiat. Isot. 45 (1994) 641-644  
(K X-ray emission probabilities)
- E. SCHÖNFELD. PTB report PTB-6.33-95-2 (1995)  
(P(K), P(L), P(M), P(N), P(O))
- E. SCHÖNFELD, H. JANSSEN. Nucl. Instrum. Methods Phys. Res. A369 (1996) 527-533  
(K-, L-shell fluorescence yields, K(beta)/K(alpha), K(alpha2)/K(alpha1), KLX/KLL, KXY/KLL)
- E. SCHÖNFELD, G. RODLOFF. PTB report PTB-6.11-98-1 (1998)  
(Auger electrons)
- E. SCHÖNFELD. Appl. Radiat. Isot. 49 (1998) 1353-1357  
(P(K), P(L), P(M), P(N), P(O))
- M. SAINATH, K. VENKATARAMANIAH. Indian J. Pure Appl. Phys. 37 (1999) 87-91  
(ICC(K), ICC(L), ICC(M) 39.75 keV)
- E. SCHÖNFELD, G. RODLOFF. PRB report PTB-6.11-1999-1 (1999)  
(X(K))
- A. KOVALIK, A.F. NOVGORODOV, E.A. YAKUSHEV, V.M. GOROZHANKIN, P. PETEV, M.A. MAHMOUD. Proc. 50th Ann. Conf. Nucl. Spectrosc. Struct. At. Nuclei, St. Petersburg, Russia (2000) 176  
(Rh-103m KLL Auger spectrum)
- A. KOVALIK, E.A. YAKUSHEV, D.V. FILOSOFOV, V.M. GOROZHANKIN, V.G. KALINNIKOV. Proc. 50th Ann. Conf. Nucl. Spectrosc. Struct. At. Nuclei, St. Petersburg, Russia (2000) 175  
(39.7537(21)-keV gamma-ray energy)
- A. KOVALIK, E.A. YAKUSHEV, D.V. FILOSOFOV, A.F. NOVGORODOV, V.M. GOROZHANKIN, I. STEKL, P. PETEV, M.A. MAHMOUD. Proc. 50th Ann. Conf. Nucl. Spectrosc. Struct. At. Nuclei, St. Petersburg, Russia (2000) 178  
(Rh-103m KL1L2 Auger spectrum)
- E. SCHÖNFELD, H. JANSSEN. Appl. Radiat. Isot. 52 (2000) 595-600  
(X-ray emission probabilities, Auger-electron emission probabilities)
- D. DE FRENNE, E. JACOBS. Nucl. Data Sheets 93 (2001) 447-598  
(Nuclear levels)
- I.M. BAND, M.B. TRZHASKOVSKAYA, C.W. NESTOR JR., P.O. TIKKANEN, S. RAMAN. At. Data Nucl. Data Tables 81 (2002) 1-334  
(Theoretical ICC)

- S. RAMAN, C.W. NESTOR JR., A. ICHIHARA, M.B. TRZHASKOVSKAYA. Phys. Rev. C 66 (2002) 044312  
(Theoretical ICC)
- T.P. BERLYAND, E.I. GRIGOR'EV, V.P. ORLOV. Measurement Techniques 45 (2002) 974-977  
(Gamma-ray emission probabilities, X-ray emission probabilities, K(alpha)/K(beta), (P(357 keV)/P(K X-ray)))
- YU.S. POPOV, L.V. ZAKHAROVA, V.N. KUPRIYANOV, O.I. ANDREEV, A.N. PAKHOMOV, F.Z. VAKHETOV. Radiochemistry 46 (2004) 209-210  
(Pd-103 half-life, gamma-ray emission probabilities, X-ray emission probabilities, K(alpha)/K(beta))
- T. KIBÉDI, T.W. BURROWS, M.B. TRZHASKOVSKAYA, P.M. DAVIDSON, C.W. NESTOR JR. Nucl. Instrum. Methods Phys. Res. A589 (2008) 202-229  
(Theoretical ICC)
- D. DE FRENNE. Nucl. Data Sheets 110 (2009) 2081-2256  
(Nuclear levels)
- F. TARKANYI, A. HERMANNE, B. KIRALY, S. TAKACS, F. DITROI, J. CSIKAI, A. FENYYESI, M.S. UDDIN, M. HAGIWARA, M. BABA, T. IDO, YU.N. SHUBIN, A.V. IGNATYUK. Appl. Radiat. Isot. 67 (2009) 1574-1581  
(Discrepancies in gamma-ray and K X-ray emission probabilities)
- K.S. KRANE. Phys. Rev. C 81 (2010) 044310  
(Gamma-ray energies, Rh-103 nuclear level energies)
- B.Q. LEE, T. KIBÉDI, A.E. STUCHBERY, K.A. ROBERTSON. Compt. Math. Methods Med. Article ID 651475 (2012) 1-14  
(Atomic radiation)
- B.Q. LEE, T. KIBÉDI, A.E. STUCHBERY, K.A. ROBERTSON, F.G. KONDEV. Proc. Heavy Ion Accelerator Symposium (HIAS2013), 8-12 April 2013, Canberra, Australia, EPJ Web of Conferences 63 (2013) 01002  
(Auger electrons)
- X. MOUGEOT. Erratum, Phys. Rev. C 92 (2015) 059902  
(Beta decay, BetaShape code)
- X. MOUGEOT. Phys. Rev. C 91 (2015) 055504  
(Beta decay, BetaShape code)
- B.Q. LEE, HOOSHANG NIKJOO, J. EKMAN, P. JONSSON, A.E. STUCHBERY, T. KIBÉDI. Int. J. Radiat. Biol. 92 (2016) 641-653  
(Atomic radiation)
- N. NICA, J.C. HARDY, V.E. IACOB, V. HORVAT, H.I. PARK, T.A. WERKE, K.J. GLENNON, C.M. FOLDEN III, V.I. SABLA, J.B. BRYANT, X.K. JAMES, M.B. TRZHASKOVSKAYA. Phys. Rev. C 98 (2018) 054321  
(ICC(K), ICC(total), mixing ratio 39.75 keV)
- J. RIFFAUD, P. CASSETTE, D. LACOUR, V. LOURENCO, I. TARTÈS, M.A. KELLETT, M. CORBEL, M.-C. LÉPY, C. DOMERGUE, C. DESTOUCHES, H. CARCREFF, O. VIGNEAU. Appl. Radiat. Isot. 134 (2018) 399-405  
(Rh-103m IT decay: absolute K X-ray emission probabilities, absolute gamma-ray emission probability)
- X. MOUGEOT. Appl. Radiat. Isot. 154 (2019) 108884  
(EC/positron decay, BetaShape code)
- B.P.E. TEE, T. KIBÉDI, B.Q. LEE, M. VOS, R. DU RIETZ, A.E. STUCHBERY. Proc. Heavy Ion Accelerator Symposium (HIAS2019), 9-13 September 2019, Canberra, Australia, EPJ Web of Conferences 232 (2020) 01006  
(Atomic radiation)
- F.G. KONDEV, MENG WANG, W.J. HUANG, S. NAIMI, G. AUDI. Chin. Phys. C 45 (2021) 030001  
(Rh-103m Q(IT))
- MENG WANG, W.J. HUANG, F.G. KONDEV, G. AUDI, S. NAIMI. Chin. Phys. C 45 (2021) 030003  
(Q-values)
- J. RIFFAUD, M.-C. LÉPY, P. CASSETTE, M. CORBEL, M.A. KELLETT, V. LOURENCO. Appl. Radiat. Isot. 167 (2021) 109298  
(Pd-103 half-life, absolute K X-ray emission probabilities, relative L X-ray emission probabilities, absolute gamma-ray emission probabilities)
- X. MOUGEOT. Appl. Radiat. Isot. 201 (2023) 111018  
(Beta decay, BetaShape code)
- B.P.E. TEE, T. KIBÉDI, B.Q. LEE, A.E. STUCHBERY. BrIccEmisDB database, in preparation for submission to At. Data Nucl. Data Tables (2023)  
(Atomic radiation)

